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THIRD SEMESTER **B.Tech. (SE/EP/EE/ME/CE/EEE)**

END SEMESTER EXAMINATION NOV.-DEC.-2011

SE/EP/EE/ME/CE/EEE-206 ENGINEERING ECONOMICS

Time: 3 Hours ***Maximum Marks : 70***

Note : Answer any *FIVE* questions.
Assume suitable missing data, if any.

1[a] Discuss characteristics of four different forms of market. 7

[b] (i) In the design of building to be constructed at Alpha state, the designer is considering the type of window frame to specify. Either steel or aluminum window frames will satisfy the design criteria. Because of the remote location of the building site and lack of building materials in alpha state, the window frames will be purchased in Beta state and transported for a distance of 2,500 Km to the site. The price of window frames of the type required is Rs1,000 each for aluminum frames . The weight of steel window frames is 75 kg each and that of aluminum widow frame is 28 Kg each. The shipping rate is Rs.1/Kg/100Km. Which design should be specified and what is the economic advantage of the selection? 3

(ii) A person is planning for his retired life. He has 10 more years of service. He would like to deposit 20% of his salary, which is Rs.4,000 at the end of the first year, and therefore he wishes to deposit the amount with an annual increase of Rs.500 for the next 9 years with an interest rate 15%. Find the total amount at the end of the 10th year of the above series. 4

2[a] How emerging areas of your branch of engineering may be applied in a slum to improve its condition and life? Discuss. 10

[b] If demand function for two products X_1 and X_2 are given by

$X_1 = P_1^{-a11} e^{a12P2a1}$

$X_2 = P_2^{-a22} e^{a22p1a2}$

Show nature of both products X_1 and X_2. 4

3[a] Suggest measures to maintain balance between economic development and environmental protection. 7

[b] (i) Three engineering graduates decide open a business, and all three devote their full time to its management. What opportunity cost would you assign to their time? 3+4

(ii) The market supply and demand function for deluxe pizzas in a small town are given by

Demand $Q = 100 - 3.5p$

Suply $Q = 15 + 1.5\,p$

(i) Determine the equilibrium price and quantity

(ii) If the city government levies a tax of Rs.30.00 /pizza on the pizza parlor, determine the new equilibrium price and quantity of pizza.

4[a] Why do businessmen such as automobile dealers tend to locate together? 7

[b] (i) A company has to replace a present facility after 15 years at an outlay of Rs.5,00,000. It plans to deposit an equal amount at the end of every year for the next 15 years at an interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years. 3

(ii) An engineer has two bids for an elevator to be installed in a new building The details of the bids for the elevator are as follows:

Bid	Engineer's Estimates		
	Initial Cost	Service life (yrs)	Annual operations & maintenance cost (Rs.)
α Elevator Inc.	4,50,000	15	27,000
β Elevator Inc.	5,40,000	15	28,500

Determine which bid should be accepted, based on the present worth method of comparison assuming 15% interest rate, compound annually. 4

5[a] Discuss Business risk in globalized market. 7

[b] (i) Two alternatives are under consideration for a tapered fastening pin. Either design will serve the purpose and will involved the same material and manufacturing cost except for the lathe and grinder operations.

Design a will require 16 hours of lathe time and 4.5 hours of grinder time per 1000 units Design B will require 7 hours of lathe time

and 12 hours of grinder time per 1000 units. The operating cost of the lathe including labour is Rs.200/hour The operating cost of the grinder including labour is Rs.150/hour which design should be adopted if 1,00,000 units are required/.year and what is the economic advantage of the best alternatives? 3

(ii) A certain individual firm desires an economic analysis to determine. Which of the two machines is attractive in a give interval of time. The minimum attractive rate of return for the firm is 15%. The following data are to be used in the analysis

	Machine X	Machine Y
First cost	1,50,000	2,40,000
Estimated cost	12years	12 years
salvage cost	0	6,000
annual maintenance cost	0	4,500

Which machine would you choose? Base your answer on annual equivalent cost. 4

6[a] Discuss role of Central Bank in a liberalized economy. 7

[b] (i) A manufacturer of motor cycles buys side boxes at Rs.240 each. In case he makes it himself, the fixed and variable costs would be Rs. 30,00000 and Rs90/side box respectively Should the manufacturer make or buy the side boxes if the demand is 2,500 side boxes? 3

(ii) An item has an yearly demand of 1,000 units The different costs with regard to make and buy are as follows. Determine the best option

Detail	Buy	Make
item cost/unit	Rs.6.00	Rs.5.90
procurement cost/order	Rs.10.00	--
set-up cost/setup	--	Rs.50.00
annual carrying cost /item /year	Rs.1.32	Rs.1.30
production rate/year		6000 units

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7[a] Give eight salient features of Indian Economy. 7

[b] (i) A person is planning a new business The initial outlay and cash flow pattern for the new business are as listed below. The expected life of the business is five years. Find the rate of return for the new business. 3

period	0	1	2	3	4	5
Cash flow (Rs.0	- 1,00,000	30,000	30,000	30,000	30,000	30,000

(ii) A small business with an initial outlay of Rs.12,000 yields Rs.10,000 during the first year of its operation and the yield increases by Rs.1,000 from its second year of operation. At the end of the life of the business, the salvage value is zero. Find the present worth of the business by assuming on interest rate of 18%, compound annually.

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